B.Sc (HONS.) IN CSE FIRST YEAR, SECOND SEMESTER EXAMINATION, 2023

STATISTICS AND PROBABILITY

[According to the New Syllabus]

Subject Code : 510227

Examination Code-5612

Time—3 hours

Full marks—80

[*N.B. The figures in the right margin indicate full marks. Answer any* **four** *questions.*]

Marks

1. *(a)* Distinguish between *(i)* Population and Sample *(ii)* Descriptive statistics and inferential statistics. 4

   (b) A study of air pollution in a city yielded the following daily reading the concentration of sulfur dioxide: 12

   40, 140, 170, 110, 180, 200, 130, 170, 100, 70, 150, 90, 100, 160, 120, 270, 100, 120, 50, 60, 80, 150, 50, 90, 110, 130, 70, 140, 90, 170.

   (i) Construct a frequency distribution table with suitable size of class interval.

   (ii) Also find relative frequency.

   (c) The following table shows the (%) of female population of Bangladesh in five divisions: 4

Division	Barisal	Chittagong	Dhaka	Khulna	Rajshahi
Populations (%)	46·3	26·2	30·0	40·2	50·5

   Construct a pie chart.

2. *(a)* Describe the various measures of central tendency. 5

   (b) For two non zero positive numbers, prove that AM ≥ GM ≥ HM. 5

   (c) The obtained marks of the seven students in a class examination are to be found 4, 5, 7, 8, 11, 13, 15. 5

   By using the above mentioned data, prove that, $\sum(x_i - \bar{x})^2 < (x_i - a)^2$; where, $a = 10$.

   $\frac{N}{\sum f/x}$

   (d) Calculate geometric mean and harmonic mean from the following frequency distribution: 5

Class	10—20	20—30	30—40	40—50	50—60
Frequency	7	10	18	6	5

[*Please turn over*

Antilog $\frac{\sum f \log x}{N}$

SD = $\sqrt{\frac{fx^2}{N} - \bar{x}}$ MD = $\frac{\Sigma f|x|}{N}$ SD

Marks

3. *(a)* What are the qualities of a good measure of dispersion? Which measure is suitable and why? 5

(b) Distinguish between central moments and raw moments. 5

(c) The following table shows the distribution of the patients care to take services in the outdoor of a hospital: 10

Age	0—15	15—30	30—45	45—60	60—75
Number of Patients	15	5	4	8	8

Determine the mean deviation and standard deviation and also verify that mean deviation is less than standard deviation.

4. *(a)* Write the difference between correlation coefficient and regression coefficient. 4

(b) Prove that correlation is equal to geometric mean of regression coefficients. 4

(c) The following figure relate to the advertisement expenditure (X) (in taka lakh) Sales (Y) (in crores taka): 12

X	25	60	48	62	66	63	42	35
Y	10	20	13	15	20	21	8	10

(i) Find correlation coefficient,

(ii) Find regression equation Y on X,

(iii) Estimate the amount of sales when advertisement expenditure is 44 lakh.

$\sqrt{\frac{\Sigma f D (X-\bar{X})}{N}}$

5. *(a)* Define intra-class and Bi-serial correlation. 4

(b) From the following results obtain the regression equations and estimate the yield of crops when the rainfall is 26 cm and rainfall when the yield is 420 kg. 7

	Yield (Y) (kg)	Rainfall (X) cm
Mean	440	28·8
Standard deviation	35·6	5·6

Correlation coefficient $\gamma = 0{\cdot}80$.

(c) If $\gamma_{12} = 0{\cdot}60$ $\gamma_{13} = 0{\cdot}50$ and $\gamma_{23} = 0{\cdot}45$ then calculate $\gamma_{12\cdot3}$, $\gamma_{13\cdot2}$ and $\gamma_{23\cdot1}$. Also find $R_{1\cdot23}^2$. 9

6. *(a)* Define box and whiskers plots. 3

(b) The Spearman's rank correlation coefficient between demand and supply is found to be 0·143. If the sum of squares of the differences in rank is given to be 48, find the value of *n*. 4

(c) Define stem and leaf plot. 3

(d) The following table gives the aptitude test scores and productivity indices of 6 workers selected at random: 10

Aptitude test score	60	62	65	70	50	53
Productivity index	68	60	62	80	85	40

Estimate: *(i)* The productivity index of a worker whose aptitude test score is 90. *(ii)* The aptitude test score of a worker whose productivity index is 70.

12.5 12.5 8 8 32 32

B.Sc. (HONS.) IN CSE FIRST YEAR SECOND SEMESTER EXAMINATION, 2021

STATISTICS AND PROBABILITY

[According to the New Syllabus]

Subject Code: 510227

Examination Code : 5612

Time—3 hours

Full marks—80

[N.B. The figures in the right margin indicate full marks. Answer any four questions.]

Marks

1. (*a*) Define Statistics. Discuss the importance and uses of Statistics. 6

   (*b*) Define data. What are the various types of data available for statistical analysis? 6

   (*c*) What is frequency distribution? The obtained marks of 30 candidates in an examination are given below : 8

34	36	31	46	76	86	42	44
32	46	40	54	66	56	50	42
33	80	77	81	46	40	60	63
64	76	56	57	37	70		

   (*i*) Construct a frequency distribution table by using suitable class interval.

   (*ii*) Construct a stem and leaf diagram for this data.

2. (*a*) Distinguish between : 5

   (*i*) Qualitative data and quantitative data.

   (*ii*) Discrete variable and continuous variable.

   (*b*) Describe various measures of central tendency. 5

   (*c*) For two non-zero positive numbers. Prove that, A.M.XH.M= $(G.M.)^2$. 5

   (*d*) Mean of 540 observations is found to be 60. If at the time of calculation two terms are wrongly taken as 77 and 67 instead of 47 and 57. Find the correct mean. 5

3. (*a*) What is questionnaire? What precautions should be taken in drafting a questionnaire? 8

   (*b*) Which measure is suitable for measures of dispersion and why? 5

   (*c*) Define classification. Describe the different types of classification. 7

[Please turn over

Marks

4. (*a*) Define measures of dispersion. Describe the characteristics of an ideal measure of dispersion. 4

(*b*) Describe the merits and demerits of mean deviation and standard deviation. 6

(*c*) Differentiate between standard deviation and coefficient of variation. 5

(*d*) The average weekly wages and standard deviations of two factories are as follows : 5

Factory	Average weekly wages	Standard deviation	No. of workers
A	460	50	100
B	490	40	80

(*i*) Which factory pays larger amount for weekly wages?

(*ii*) Which factory shows larger variability in the distribution of wages?

5. (*a*) What is Kurtosis? Explain the different kinds of Kurtosis with the help of graph. 5

(*b*) Following are the heights and weights of 8 students of B.Sc. (Hons.) class : 5

Height (inch)	62	72	68	64	70	66	64	72
Weight (kg)	50	65	64	56	62	63	60	64

Draw a scatter diagram and indicate whether the correlation is positive or negative.

(*c*) Karl Pearson's coefficient of correlation between two variates x and y is 0·75, standard deviation of x is 3 and their covariance is 8·2. Find the variance of y. 5

(*d*) The worker's age are given below : 5

Age (year) : 25, 27, 30, 31, 33.

Determine the first four central moments.

Marks

6. (*a*) Define multiple and partial correlations. 5

(*b*) Discuss the different nature of correlation with the help of scatter diagram. 5

(*c*) Calculate Karl Pearson's coefficient of correlation from the following data : 5

x	10	12	21	4	7
y	8	7	17	2	6

(*d*) If the standard deviation of variable x is 6 and the two regression equations are $12x - 15y + 99 = 0$ and $60x - 27y = 321$. 5

Find the average value of x and y.

B.Sc. (HONS.) IN COMPUTER SCIENCE AND ENGINEERING
FIRST YEAR, SECOND SEMESTER EXAMINATION, 2020
STATISTICS AND PROBABILITY

[According to the New Syllabus]
Subject Code: CSE-510227
Examination Code: 5612
Time 3 hours Full marks-80

[N.B. The figures in the right margin indicate full marks, Answer any four questions.]

1. (a) Define statistics, population, sample. 3
(b) Describe the characteristics of statistics. 6
(c) Distinguish between primary and secondary data. 5
(d) With a suitable example briefly explain the various methods of scales of measurement. 6

2. (a) What do mean by frequency distribution? Explain the different types of frequency curve. 5
(b)What is variable? Explain various kinds of variable with example. 5
(c) The daily wages taka of 40 workers are given below : 10
88 203 108 68 101 129 119 149 210 103 96 104 146 127 97 89
123 108 94 149 93 118 136 187- 102- 191 148 163 172 93 104 132
131 142 92 87 144 183 105 195
(i) Construct a continuous frequency distribution table with suitable size of class interval.
(ii) Draw a frequency curve and polygon. 6
(iii) Estimate the limits of wages of central 50% of the workers.

3. (a) Define and explain central tendency. 3
(b)What are the essential characteristics of an ideal average? Which measure of central tendency is the best and why? 5
(c) Prove that for two non-zero positive values AM > GM >_ HM. 6
- The average income of a male employee in a firm is Tk. 1,000 and that of female is Tk. 800. But the mean income of employee is Tk. 920. Find the percentage of male and female employee. 6

4. (a) Distinguish between absolute and relative measures of dispersion.
(b) The amounts of blood pressure of a patients seen for a sample of days last year were : 103, 97, 101, 106 and 103. Determine the mean deviation. 5
(c) What is skewness? Explain different, kinds of skewness with the help of graph. 6
(d) A and B are two factory. The average weekly wages and standard deviation of wages of the workers are given below : 4
-Factory Average weekly Standard- Number of Wages (Tk.)
Deviation workers
A 1560 90 200 B 1580 1 70 160
Calculate the combined mean and standard deviation of wages of

the whole workers in the two factory.

5. (a) Distinguish between central moments and raw moments.

(b) Briefly express the relationship between the raw moments and central moments. 7

(c) What is rank correlation? Why we need rank correlation? 3

(d) Calculate the rank correlation co-efficient from the following data on hourly sales (X) and expenses (Y) of 10 stores. 5

X	50	50	55	60	65	65	65	60	60	60
Y	11	13	14	16	12	19	20	15	17	18

6. (a) Distinguish between correlation and regression. 5

(b) Regression corpff cient is not a pure number-Explain. 5

(c) How do you estimate the regression line? 5

(d) By using the following data : 5

X	30	45	32	28	27	24
Y	28	40	27	24	26	18

Estimate the regression equation of Y on X.

B.Sc. (HONS) IN COMPUTER SCIENCE AND ENGINEERING
FIRST YEAR SECOND SEMESTER EXAMINATION, 2019

According to the.Vew Syllabus

CSE-510227

(Statistics and Probability)

Time-3 hours

Ful t m arks-80

[N.B. The figures in the right margin indicate full marks. Answer any four questions.]

1. (a) Define statistics. Discuss the importance of statistics

(b) What do you mean by statistical data? Describe the method of primary data collection.

(c) Distinguish between:

(i) Discrete variable and continuous variable.

(ii) Qualitative variable and quantitative variable.

(d) Describe frequency polygon and cumulative frequency curve or ogives.

2. (a) What are the differences between histogram and bar diagram?

(b) In a class examination, the marks obtained bv 30 students are given below: 44, 32, 36, 56, 50, 34, 31, 46, 86, 76, 42,46, 40. 56, 66, 42, 33, 80, 79, 81, 49, 40, 60. 63. 64, 76, 56, 57, 70, 82

(i) Construct a frequency distribution table by using suitable class interval.

(ii) Fit-Construct a-stem and leaf diagram for this data.

(c) Define the following terms:

(i) Arithmetic Mean

(ii) Median

(iii) Mode

3. (a) Do-scribe the properties of arithmetic mean.

(b) For two non-zero positive numbers, prove that, AM X HM = (GM) 2

(c) Define variance. What are the qualities of good measure of dispersion?

Which measure is suitable and why?

(d) The mean salary paid to 200 employees in a shoe factory was found to be 250 0. Later on, it was discovered that the salaries of two employees were wrongly taken as 3.000 and 3.500 instead of 3.500 and 3,200. Commute the correct mean salary.

4. (a) What do you mean by kurtosis? Explain the different types of kurtoses with the help of diagram.

(b) Describe the absolute measures of dispersion.

(c) The first four moments of a distribution about the value 5 are 2, 20, 40 and 50 respectively. Obtain the first four central moments. B1. and B2 Comment on the skewness and kurtosis of the distribution.

(d) Determine the standard deviation from the obtained marks of the CSE students.:. .1.6,12; 14, 1,5, 18.

5. (a) Describe the different types of simple correlation.

(b) Prove that the coefficient of correlation is an independent of origin and scale.

(c) find

(d) the correlation coefficient from-the following information's: $\sum 2: = 56$, $\sum y = 40$, Karl Pearson's coefficient of correlation between two_ variates x and y is + 0. 75. Standard deviation of x is 3 and their covariance is 8.2. Find the variance of y.

6. (a) What do you mean by regression analysis? State the important properties of regression coefficient.

(b) By using the following information's: Find the regression coefficient of yon x.

(c) Given the following information:

	Advertising Sales expenditure (Tk. in lac)	Sales(Tk. In crore)
Mean	35	82
Standard deviation	9.03	17.16

(i) Calculate the two regression equations.

(ii) Estimate the likely sales for an advertising expenditure of Tk: 40 lac.

(iii) What should be the advertising expenditure for attaining sales target Tk. 90 crore?

B. Sc (HONS.) IN CSE PART-I, SECOND SEMESTER EXAMINATION, 2018

[According to the New Syllabus]

CSE-510212

Examination Code : 5612

(Statistics end Probability)

1. (a) Define statistics. Discuss the characteristics of statistics. 4

(b) What are the differences between primary data and secondary data? 4

(c) What is variable? Describe various kinds of variable with example. 6

(d) Define scale of measurement. Discuss the various type of scale of measurement. 6

2. (a) What is frequency distribution? Discuss the different steps of construction of frequency distribution. 5

(b) The daily- wages (Taka) of 50 workers are given below : 10

160	88	108	204	139	68	129	101	149	119

132	103	210	96	104	127	146	89	97	170
120	187	123	136	108	118	94	149	93	175
131	191	102	148	163	172	93	104	132	135
125	195	105	144	187	92	131	136	142	128

(i) Construct a continuous frequency distribution with suitable size of class interval.

(ii) Draw a histogram and find mode from histogram.

(c) Discuss the different measures of central tendency.

3. (a) Define median and mode with uses. 5

(b) Prove that for two non-zero positive values AM ≥ GM ≥ HM. 5

(c) Calculate Arithmetic mean, Geometric mean and Harmonic mean from the following frequency distribution: 10

Daily sales (Thousand Tk.)	20-30	30-40	10-50	50-60	60-70	70-80
Number of shops	4	7	16	12	6	5

4. (a) What is skewness? Describe the different frequency curves based on skewness. 3

(b) Distinguish between absolute and relative measures of dispersion. 5

(c) If the obtained marks of 5 students (out of 20) are 8, 7, 6, 3, 12, then prove that $100\sqrt{n-1} > CV$. 4

(d) Calculate the quartile deviation from the following information : 8

Daily Income (in Tk.)	50	60	70	80	90	100	110	120
Number of workers	10	12	14	16	20	8	4	2

Also develop a box plot.

5. (a) What is scatter diagram? Discuss the different nature of correlation with the help of scatter diagram. 5

(b) Prove that the value of coefficient of correlation lies between 1- and – 1. 5

(c) Calculate Arithmetic mean and standard deviation for n natural numbers. 5

(d) Calculate the rank correlation coefficient for the following data giving ranks awarded by two judges to 10 participants in a musical contest : 5

Rank by Judge I	3	5	4	8	9	7	1	2	6	10
Rank by Judge II	4	6	3	9	10	7	2	1	5	8

6. (a) Distinguish between correlation and regression. 5

(b) If the standard deviation of variable x is 6 and the two regression equations are $12x - 15y + 99 = 0$, $60x - 27y = 321$. Calculate (i) the average value of x and y(ii) correlation coefficient (iii) variance of y. 8

(c) Observe the following information : 7

			Boys
		Intelligent	Un-intelligent
Father	Skilled	40	30
	Unskilled	70	54

Do these figures support the hypothesis that skilled fathers ha,, a intelligent boys?

B.Sc (HONS.) IN CSE PART-I, SECOND SEMESTER EXAMINATION, 2017
CSE-.127
(STATISTICS AND PROBABILITY)
Examination Code : 612

1. (a) Define data. Write down the difference Between primary data and secondary data. 5
(b) Discuss the importance and uses of statistics. 5
(c) In a class examination the marks obtained by 30 students are given below: 10

32	33	55	47	21	50	27	26	24	33
62	42	38	15	45	32	44	48	68	49
40	52	30	17	44	58	48	39	42	37

(i) Construct a frequency distribution;
(ii) Draw an ogive curve;
(iii) Obtain the range of marks middle 50%.

2. (a) State situation to use measure of central tendency. 4
(b) What do you mean by Arithmetic Mean, Median and Mode? 6
(c) The average income of male employee in a firm is Tk. 1,000 and that of female is Tk. 800. But the mean income of the employees is Tk. 920. Find the percentage of male and female employees. 5
(d) Prove the $\log G = \dfrac{n_1 \log G_1 + n_2 \log G_2}{n_1 + n_2}$

3. (a) Distinguish between absolute and relative measures of dispersion. 5
(b) Prove that, for a series of n observations, standard deviation is not less than the mean deviation from mean. 5
(c) An organization has two units A and B. An analysis of daily wages paid to workers gave the following results:

	Unit-A	Unit-B
No. of workers	600	700
Average daily wages	120	170
Standard deviation	0	6

(i) Which unit pays larger amount of weekly wage?
(ii) In which unit there is greater variability in wage distribution?
(iii) Find the combined mean wage and the combined standard deviation wage for the whole organization.

4. (a) Briefly express the relationships between the raw moments and central moments. 4
(b) What is mathematically expectation? Write down the properties of expectation. 5
(c) Write down the properties of regression co-efficient. How do you estimate regression co-efficient from regression line?
(d) What do you mean by probability distribution? Give examples. Mathematically prove

that Arithmetic Mean and Mathematical Expectations are same value.

5. (a) Define probability. Write the difference between priori probability and posterior probability. 5

(b) State and prove Bayes' theorem. 5

(c) Prove that, the value of probability lies between 0 to 1. 2

ie, $0 \le P(A) \le 1$.

(d) In a city male and female each from 50% of the population. it is known that 20% of the males and 5% of the females are unemployed. A student studying the employment situation selects an unemployed person at random. Which is the probability that the person so selected is (i) male, (ii) female?

6. (a) Define Bemouli Distribution. A probability distribution is given below:

X	0	1	2
P(x)	25	50	25

Compute the expected value of (I) X3 and (ii) 2X.

(b) What is random variable and variance of a random variable? 4

(c) Define the following terms with example: 5

(i) Event;

(ii) Independent Event;

(iii) Dependent Event;

(iv) Sample Space.

(d) The mean and variance of Binomial distribution are 4 and $\frac{4}{3}$ respectively.

Find (i) Probability Function,

(ii) $P(X \ge 1)$.

B.Sc (HONS.) IN CSE, PART-I SECOND SEMESTER EXAMINATION, 2016

CSE-126

(Statistics and Probability)

Examination Code : 612

Time—3 hours

Full marks—80

1. (a) Define variable with example. How does a discrete variable differ from a continuous variable? 5

(b) What is frequency distribution? Write different steps of construction of a frequency distribution. 5

(c) In a class examination, the marks obtained by 30 students are given below:— 10

40, 75, 35, 54, 42, 80, 09, 28, 72, 88
82, 03, 60, 40, 17, 32, 41, 70, 57, 30
41, 33, 40, 12, 17, 18, 72, 65, 30, 31.

(i) Taking class 0—10 construct a frequency distribution table from the above data;

(ii) Draw a histogram with the help of data.

2. (a) Describe the various measures of central tendency. 5

(b) For two non-zero positive values. Prove that, AM > GM > HM. 5

(c) Calculate mean, median and mode from the following frequency distribution :— 10

Obtained marks	40-50	50-60	60-70	70-80	80-90
Number of students	15	20	35	20	10

3. (a) What is coefficient of variation? Why coefficient of variation is called best measurement? 5

(b) The scores of two circuit players for 1 0 innings are given below : — 10

Cricketer A	101	40	0	13	67	78	6	90	0	3
Cricketer B	20	25	44	44	43	65	55	39	47	27

compare the efficiency of the two players.

(c) If x and s denote respectively mean and standard deviation of a set of non-negative quantities, then prove that, $\bar{x}\sqrt{b-1} \geq s$.

4. (a) What is random variable? How many types of random variables are there? 5

(b) The probability density function of a random variable x is given below: 5

$f(x) = kx; \quad 0 < r < 4$

$\quad = 0 \ ; \quad$ elsewhere

(i) Determine the value of K.

(ii) Find out the values of $p(1 < x < 2)$ and $p(x > 2)$.

(c) In a community the probability that a newly born child will be a girl is 04. Among 4 newly born children in that community, what is the probability that — 10

(i) all the four are girls

(ii) at least 2 are girls

(iii) no girl

(iv) exactly one girl

(v) at most 2 girls.

5. (a) What is probability distribution? When a distribution called discrete probability distribution? 5

(b) When binomial distribution tends to Poisson distribution. Write the characteristics of normal distribution. 5

(c) Given the following joint density function of the random variables X and Y:— 10

$$f(x, y) = \frac{x(3y^2+1)}{4} \, 0 < x < 2, 0 < y < 1 = 0 \text{ elsewhere}$$

(i) Show that this function is joint density function.

(ii) Find he marginal density function of X and 7.

6. (a) State and prove the additive law of probability with statement for two mutually disjoint events. 5

(b) Discuss the various definitions of probability with limitations. 5

(c) A box contains 8 red, 3 blue and 9 green balls. If 3 balls are drawn at random, determine the probability that : 10

(i) all 3 are red

(ii) all 3 are blue

(iii) at least 1 is blue]

(iv) 2 are red and 1 green 8^^

(v) 1 of each color.

B.Sc (HONS.) IN CSE PART-I, SECOND SEMESTER EXAMINATION, 2015

CSE-126

Examination Code: 612

(Statistics and Probability)

Time—3 hours

Full marks—80

1. (a) What are the differences between primary data any second data? 5

(b) Describe the different steps in the construction of frequency distribution. 5

(c) Describe the importance and uses of statistics. 5

(d) The daily wages (Taka) of 40 workers are given below:- 5

88	203	108	68	101	129	119	149	210	103
96	104	146	127	97	89	123	108	94	149
93	118	136	187	102	191	148	163	172	93
104	132	131	142	92	87	144	183	105	195

Construct a continuous frequency distribution with suitable size of class interval.

2. (a) What is Kurtosis? Illustrate various types of Kurtosis. 4

(b) Define geometric mean for group and ungroup data. 2

(c) Prove that, $\log G = \dfrac{n_1 \log G_1 + n_2 \log G_2}{n_1 + n_2}$ 4

(d) What do you mean by location and what are they? 2

(e) Calculate Mean, Median, Mode and 3rd quartile from the following data of some number of people and their age range:— 8

Age	Frequency
24.5—29.5	3
29.5—34.5	9
34.5—39.5	15
39.5—44.5	12
44.5—49.5	7
49.5—54.5	4

3. (a) Distinguish between absolute and relative measures of dispersion. 5

(b) What are the qualities of a good measure of dispersion? 5
Which measure is suitable and why?

(c) A and B are two factory. The average weekly wages and standard deviation of wages of the workers are given below:— 10

Factory	Average weekly wages (Tk.)	Standard deviation	Number of workers
A	1560	90	200
B	1580	70	160

Calculate the combined mean and standard deviation of wages of the whole workers in the two factory.

4. (a) What do you mean by endogenous and exogenous variable? Write with equation. 4

(b) Define :— 4

(i) Scatter; (ii) Regression; (iii) Co-efficient correlation and

(iv) Rank correlation.

(c) What are the significant of $-1 \leq r \leq +1$. 3

(d) State the properties of regression model.

(e) The given data are showing test score and salesman on an intelligence test: 7

Test score :	40	70	50	60	80	50	90	40	60	60
Sales (000Tk)	2.5	6	4	5	4	2.5	5.5	3	4.5	3

(i) Compute the regression line of sales on test score.

(ii) If a sales man makes score 90 and 100 than what are sales volume?

5. (a) Define :— 4

(i) Complementary event;

(ii) Sample space;

(iii) Mutually exclusive event;

(iv) Random experiment.

(b) State and prove Multiplicative law for two dependent events. 5

(c) A Picnic party is conducting a raffle draw from 50 tickets4 one per participant. There are 3 prizes to be awarded. If the 4 organizers of the raffle each buys one ticket, what is the probability that 4 organizers — 4

(a) win all the prizes

(b) win exactly 2 of the prizes

(c) win one of the prizes

(d) win no prize?

(d) What is mathematical expection? State expected value for two random variable.

(e) Given the following joint density function of x and y :— 4

$f(x,y) = 4xy, 0 < x < 1, 0 < y < 1$

$= 0$

Obtain E(x), E(y), E(xy) and E(x + y).

6. (a) Define binomial distribution. What are the properties of a binomial distribution? 1+4=5

(b) Prove that the mean and variance of a Poisson distribution is same. 5

(c) An unbiased coin is tossed 10 times. Find the probability of getting (i) 3 heads (ii) at least 1 head (iii) at most 1 head. 10

B.Sc (HONS.) IN CSE PART-I, SECOND SEMESTER EXAMINATION, 2014

CSE-126

Examination Code: 612

(Statistics and Probability)

Time—3 hours

1. (a) Define Statistics. Write down the characteristics of statistics. 1+4=5

(b) What is variable? Define various kinds of variables with example. 1+4=5

(c) The following table shows the wages (taka) of 40 workers :— 10

Daily wages	60-80	80-100	100-120	120-140	140-160	160-180	180-200	200-220	Total l
No. of Workers	1	9	10	6	6	2	4	2	N = 40

(i) Find frequency density & relative frequency.

(ii) Draw an ogive curve and find median and 3rd quartile.

(iii) Find the no. of workers whose wages are below 125, from the ogive curve.

2. (a) What do you mean by measures of central tendency? 2
   (b) Establish the relationship between arithmetic mean, geometric mean and harmonic mean. 4
   (c) The mean salary paid to 200 employees in a shoe factory was found to be Tk. 2800. Later on, it was discovered that the salaries of two employees were wrongly taken as 3,200 and 3,600 instead of 3,400 and 3,000. Compute the correct mean salary. 10
   (d) Which measure is superior for measure of central tendency and why? 4
3. (a) What is dispersion? Describe the absolute measures of dispersion with merits and demerits. 1+5=6
   (b) Prices of a particular commodity in five years of two cities are given below:— 10

City A (Tk.)	20	24	18	24	17
City B (Tk.)	22	23	19	13	15

Find which city had more stable prices.

   (c) Prove that, $100\sqrt{n-1} > c.v$ (for n positive observation). 4
4. (a) What is mathematical expectation? Write down the properties of expectation. 4
   (b) Define binomial distribution. Prove that the sum of all probabilities of binomial distribution is one. 6
   (c) The mean and variance of a binomial distribution are 4 and $\frac{4}{3}$ respectively. Find
      (i) Probability function;
      (ii) $p(x = 0)$;
      (iii) $p(x \geq 1)$.
   (d) The joint probability density function of x and y is given by— 6

$$f(x, y) = \frac{1}{8}(x + 8) \quad 0 \leq x \leq 2; \qquad 0 \leq y \leq 2$$
$$= 0 \text{ otherwise,}$$

      (i) Verify that it is a joint density function;
      (ii) Find the marginal density of x.
5. (a) What do you mean by probability? Write down the difference between a priori and a posterior definitions of probability. 4
   (b) Define the following terms with example :— 4
      (i) Event;
      (ii) Independent event;
      (iii) Dependent event;
      (iv) Sample space.

(c) A candidate is selected for interview for three posts. For the first post there are 3 candidates, for the second post there are 4 and for the third post there are 2 candidates. What are his chances getting at least one post?

(d) For any two non-mutually exclusive events A and B, prove that
$P(A \cap B) = P(A) - P(A \cap B)$.

6. (a) The probability of density function of a random variable x is given, below :
$f(x) = \frac{x}{8}; 0 < x < 4.$ 4
Find the value of $E(2x^2 + 3)$.

(b) State the properties of binomial distribution. Also prove that the sum of all probabilities of binomial distribution is one. 6

(c) The mean of a binomial distribution is 40 and standard deviation 6. Calculate n, p and q. 4

(d) The probability that Bangladesh win a Cricket -Test match against Pakistan is given by $\frac{1}{3}$ If Bangladesh and Pakistan play three Test matches, use binomial distribution to find the probability that —
(i) Bangladesh will loss ail three Test matches;
(ii) Bangladesh will win at least one Test match;
(iii) Bangladesh will win all three Test matches.

BSc (HONS). IN CSE PART-I, SECOND SEMESTER EXAMINATION, 2013

CSE-126

Examination Code : 612

(Statistics and Probabillity)

Time—3 hours

Full marks— 80

1. (a) Define date. How are they collected? What are the various types of data available for statistical analysis? 6

(b) What is frequency distribution? The daily wages (Taka) of 40 workers are given below:-

86	204	108	68	101	129	119	149	210
103	96	104	146	127	97	89	123	108
91	149	93	118	136	187	102	191	148
163	172	93	104	132	131	142	92	87
144								

(i) Construct a continuous frequency distribution;
(ii) Draw Histogram and Relative frequency polygon. 4

(c) What is frequency curve? Explain different types of frequency curves. 3

2. (a) State the situation to use the measure of central tendency

(b) From the following data find out the missing frequency when mean is 22:—

Central value :	10	15	20	25	30
Frequency :	5	10	154	7	5

(c) The average monthly income of male employees in a firm is Tk. 1,000 and that of female is Tk. 800. But the mean income of all the employees is 920. Find the percentage of male and female employees.

(d) What is statistical average? Mention different types of averages. State the characteristics of arithmetic mean.

3. (a) Differentiate between standard deviation and co-efficient of variation. 4

(b) Find the mean deviation and standard deviation of the following data:- 10

Monthly Income (In thousand)Tk.	No. of families
5—9	15
10—14	30
15—19	55
20—24	17
25—29	10
30—34	3

(c) Why standard deviation is considered to be the best measure? State the some uses of standard deviation. 6

4. (a) Write the difference between correlation co-efficient and regression co-efficient. 6

(b) Prove that the unit of correlation of co-efficient os-1 to 1. 3

(c) What is least square method? How you calculate the regression parameters with the help of least square method? 4

(d) The following figure relate to the advertisement expenditure (X) (in lakh taka) sales (Y) (in crores taka):- 10

X	55	61	60	64	48	62	66	63
Y	12	15	14	18	13	16	20	21

(i) Find correlation co-efficient;

(ii) Find regression equation X on Y.

5. (a) What is probability distribution? Define binomial and poisson distribution. 6

(b) A continuous random varible x has the following density function 8

$f(x) = \frac{1}{8} x$ for $0 < x < 4$:-

(i) Verify that is satisfies the conditions $\int_{-\infty}^{+\infty} f(x)dx = 1$;

(ii) Find $F(0 < x < 3)$;

(iii) Find $F(x < 4)$.

(c) Let X and Y have the following distribution $f(x, y) = 2(x + y - 2xy)$ for $0 < x < 1, 0 < y < 1$. Find E (x), E(X + y), E (x) 6

6. (a) Define probability. If A and B are two independent events then prove that $\overline{A}$ and $\overline{B}$ independent. 5

(b) A box contains 4 red, 5 white and 6 black balls. 4 balls are drawn at random from the box. Find the probability that among the balls drawn there is at least one ball of each colour. 6

(c) Define marginal and conditional probability with examples. 4

(d) Let x and y have the following distribution $f(x, y) = 2(x + y - 2xy)$; $0 \le x \le 1, 0 \le y \le 1 = 0$; else where:- 5

(i) Check that $f(x, y)$ is a density;

(ii) Find marginal density function;

(iii) Determine the conditional probability function of x given y.